🤟 Sign Language Communication

Training Manual for Healthcare Staff & Patients

Tygerberg Hospital — Deaf AI Healthcare Assistant

Version 2.0 · Powered by Google MediaPipe ML + Custom Medical Gestures

📋 Table of Contents

1. System Overview
2. How the System Detects Signs
3. Standard SASL Gestures (Universal)
4. Technique: How to Sign Clearly
5. Quick Phrase Boards by Department
6. Building Sentences
7. Detection Modes: Words, Letters, Numbers
8. Troubleshooting
9. Practice Exercises

1. System Overview

The Deaf AI Healthcare Assistant translates hand gestures into medical words in real-time using your camera. It uses **two AI models together**:

Model	Type	What It Detects	Accuracy
Google GestureRecognizer ML MODEL	Pre-trained neural network	7 core gestures: Open Palm, Fist, Thumbs Up, Thumbs Down, Peace/Victory, Pointing Up, ILY	High
Custom Geometric Classifier RULE-BASED	Finger angle analysis	8 extra medical gestures: Pinch, OK Sign, Horns, Call Me, Three Up, L-Shape, Palm Down, Four Up	Good

The ML model runs **first** as the primary detector. The geometric classifier adds extra medical gesture support that the ML model doesn't cover.

2. How the System Detects Signs

Detection Pipeline

1. **Camera captures your hand** — Keep your hand clearly visible, well-lit
2. **ML model checks for gesture** — Google's neural network analyzes the image
3. **Geometric classifier runs** — Checks for additional medical gestures the ML model doesn't know
4. **Stability check** — You must hold the gesture steady for ~0.4 seconds (2 frames) before it registers
5. **Word appears on screen** — The gesture is translated to a standard SASL medical word (same in every department)

> 💡 **Anti-Repetition:** The same word will NOT repeat. To say the same word again, you must first make a *different* gesture, or put your hand down and raise it again. A word cannot re-appear within 5 seconds.

3. Standard SASL Gestures (Universal)

These 15 gestures work the **same in every department**. Each gesture always means the same word — following South African Sign Language (SASL) conventions. No confusion, no department-specific overrides.

🤖 ML-Detected Gestures (Primary — High Accuracy)

	Gesture	How to Form It	Standard Word
🖐	**Open Palm** ML	All 5 fingers extended, palm facing camera, fingers spread apart	Hello
✊	**Closed Fist** ML	All fingers curled tight into palm, thumb wrapped over fingers	No
👍	**Thumbs Up** ML	Fist with thumb pointing straight up	Yes
👎	**Thumbs Down** ML	Fist with thumb pointing straight down	Feeling bad
✌	**Victory / Peace** ML	Index and middle fingers raised in a V, other fingers curled	Thank you
☝	**Pointing Up** ML	Index finger extended straight up, all other fingers curled	There
🤟	**I Love You (ILY)** ML	Thumb, index finger, and pinky extended; middle and ring curled	I love you

🔬 Geometric-Detected Gestures (Additional Medical)

	Gesture	How to Form It	Standard Word
🤏	**Pinch** GEO	Thumb and index finger touching tips, other fingers relaxed/open	Pain
👌	**OK Sign** GEO	Thumb and index form a circle, other 3 fingers extended straight	OK

🤘	**Horns** GEO	Index and pinky extended, middle and ring curled, thumb curled	Help
🤙	**Call Me** GEO	Thumb and pinky extended, other 3 fingers curled into palm	Call
🖖	**Three Up** GEO	Index, middle, and ring fingers raised; pinky and thumb curled	Medicine
👆	**L-Shape (Thumb+Index)** GEO	Thumb and index extended at an L-angle, other 3 fingers curled	Need
🖐	**Palm Down (Flat)** GEO	All fingers extended, palm facing *downward* (flat horizontal)	Please wait
🖐	**Four Up** GEO	Index, middle, ring, and pinky raised; thumb curled across palm	Wait

4. Technique: How to Sign Clearly

✅ DO

🎯	**Hold each gesture for at least 1 second** — the system needs 2 consecutive frames (~0.4s) to register
💡	**Use good lighting** — face a window or lamp; avoid backlight behind your hand
📏	**Keep hand 30–60 cm from camera** — not too close, not too far
🖐	**Exaggerate finger positions** — spread fingers wide for open palm; curl tightly for fist
⏸	**Pause between gestures** — drop hand briefly or make a neutral position between words
🎨	**Plain background** — avoid busy patterns behind your hand

❌ DON'T

🚫	**Don't switch gestures too quickly** — transitioning fast causes false detections
🚫	**Don't wave your hand around** — keep it steady while holding a gesture
🚫	**Don't show both hands** — the system tracks one hand only
🚫	**Don't hold the same gesture expecting repeated words** — each word requires: gesture → pause → gesture

> ⚠️ **To say the same word twice:** Show gesture → put hand down (or make a different gesture) → show original gesture again. Simply holding a gesture will NOT repeat the word.

5. Universal Gesture Reference

All gestures follow **South African Sign Language (SASL) standards**. Each gesture always means the **same word**, no matter which department the patient is in. This prevents confusion — a fist always means "No", a pinch always means "Pain", etc.

🤟 Complete Gesture → Word Chart (All Departments)

Gesture	Hand Shape	Always Means	Detection
🖐 Open Palm	All 5 fingers spread open	**Hello**	ML
👍 Thumbs Up	Fist with thumb up	**Yes**	ML
👎 Thumbs Down	Fist with thumb down	**Feeling bad**	ML
✊ Fist	All fingers curled tight	**No**	ML
✌ Peace	Index + middle V-sign	**Thank you**	ML
☝ Pointing	Index finger up	**There**	ML
🤟 ILY	Thumb + index + pinky out	**I love you**	ML
🤌 Pinch	Thumb + index tips touching	**Pain**	GEO
👌 OK Sign	Thumb + index circle, 3 fingers up	**OK**	GEO
🤘 Horns	Index + pinky up, others down	**Help**	GEO
🤙 Call Me	Thumb + pinky out, others curled	**Call**	GEO
🖖 Three Up	Index + middle + ring up	**Medicine**	GEO
👆 L-Shape	Thumb + index in L-angle	**Need**	GEO
🖐 Palm Down	All fingers flat, palm facing down	**Please wait**	GEO
🖐 Four Up	4 fingers up, no thumb	**Wait**	GEO

💡 **How to communicate specific symptoms:** Use the standard gestures to build sentences, then use the **Quick Phrase Board** (Section 6) for department-specific medical terms. For example: sign "Pain" + "There" (point at chest) + tap "Cannot breathe" quick phrase.

📝 Example Conversations Using Standard Gestures

Pulmonology visit:

Open Palm → "Hello" → *Pinch* → "Pain" → *Pointing* → "There" → *Horns* → "Help" → tap **"Cannot breathe"** quick phrase

→ Reads: **"Hello. Pain. There. Help. Cannot breathe."**

Ophthalmology visit:

Open Palm → "Hello" → *Pinch* → "Pain" → *Thumbs Down* → "Feeling bad" → tap **"Blurry vision"** + **"Need glasses"** quick phrases

→ Reads: **"Hello. Pain. Feeling bad. Blurry vision. Need glasses."**

Gastroenterology visit:

Open Palm → "Hello" → *Pinch* → "Pain" → *Fist* → "No" → *Three Up* → "Medicine" → tap **"Vomiting"** quick phrase

→ Reads: **"Hello. Pain. No. Medicine. Vomiting."**

6. Quick Phrase Boards

Each department also has a **Quick Phrase Board** — tap-to-send buttons for common sentences. These do NOT require gestures; the patient just taps the phrase on screen.

👁️ Ophthalmology Quick Phrases

- 👁️ Eye pain
- 😵 Blurry vision
- 💧 Eyes watering
- 🔴 Eye is red
- 👓 Need glasses
- 🌟 Sensitive to light
- 💊 Need eye drops
- 🔍 Cannot see far
- 📖 Cannot read
- 🤕 Headache from eyes
- ✨ Seeing flashes
- ⬇️ Vision getting worse

> 💡 **Tip:** Quick phrases are fastest for common complaints. Use gestures for follow-up details and conversation.

7. Building Sentences

Words detected from gestures are added to a **sentence builder** on screen. The patient can review and send the complete sentence.

Example: Hospital Visit

Step	Gesture	Word Added	Sentence So Far
1	🖐 Open Palm	Hello	Hello
2	🤏 Pinch	Pain	Hello Pain
3	☝️ Pointing	There	Hello Pain There
4	👎 Thumbs Down	Feeling bad	Hello Pain There Feeling bad
5	🤟 Three Up	Medicine	Hello Pain There Feeling bad Medicine

> 💡 **Flow:** Sign gestures one at a time → Review sentence on screen → Tap "Send" when complete → Doctor receives the message

Editing

If a wrong word appears, tap the ✕ next to it in the sentence builder to remove it before sending.

8. Detection Modes

The translator supports three modes, switchable via buttons:

Mode	Button	Use For	How It Works
Words (Glosses)	🤟 Words	Main communication — medical phrases	Shows gestures → standard SASL words. Uses ML + geometric models.
Letters	🔤 ABC	Spelling names, medications, specific words	ASL fingerspelling A–Z. Requires holding each letter 0.8s.
Numbers	🔢 123	Pain scale (1–10), room numbers, quantities	ASL number signs 0–9.

When to Use Each Mode

- **Start with Words mode** — for symptoms, requests, responses
- **Switch to Letters** — to spell a medication name, doctor's name, or unfamiliar word
- **Switch to Numbers** — when asked "rate your pain 1–10" or for dosages

9. Troubleshooting

Problem	Solution
Gesture not detected	Better lighting, hold steadier, keep hand 30–60 cm from camera
Wrong word detected	Exaggerate the gesture more — spread fingers wider, curl tighter
System shows "Collecting..."	Your gesture hasn't been stable for 2 frames yet — hold still
Same word won't repeat	Put hand down, wait 1 second, show gesture again
"No hand" message	Move hand into the camera frame; ensure good lighting
Gesture words too general	Use Quick Phrase buttons for specific symptoms (e.g. "Cannot breathe", "Blurry vision")
Camera not working	Allow camera permissions in browser, check if another app is using the camera

10. Practice Exercises

Use these exercises to train staff and patients on using the system effectively.

Exercise 1: Basic Communication (All Departments)

Practice these gestures in order, holding each for 1–2 seconds with a brief pause between:

#	Gesture	Expected Word	Hold Time
1	🖐 Open Palm	Hello	1.5s
2	— put hand down —	(reset)	1s
3	👍 Thumbs Up	Yes	1.5s
4	— put hand down —	(reset)	1s
5	✊ Fist	No	1.5s
6	— put hand down —	(reset)	1s
7	✌ Peace	Thank you	1.5s

Expected sentence: "Hello Yes No Thank you"

Exercise 2: Ophthalmology Visit

Simulate a patient describing eye problems:

#	Gesture	Expected Word
1	🖐 Open Palm	Hello
2	🤏 Pinch	Eye pain
3	☝ Pointing	This eye
4	👎 Thumbs Down	Blurry vision
5	👆 L-Shape	Need glasses
6	✌ Peace	Thank you

Expected sentence: "Hello Eye pain This eye Blurry vision Need glasses Thank you"

Exercise 3: Emergency Scenario

#	Gesture	Expected Word
1	🤘 Horns	Emergency
2	🤏 Pinch	Severe pain
3	☝️ Pointing	Bleeding here
4	👎 Thumbs Down	Getting worse
5	🤟 Three Up	Accident

Expected sentence: "Emergency Severe pain Bleeding here Getting worse Accident"

Exercise 4: Speed Test

Once comfortable, try completing Exercise 2 in under 30 seconds. Remember:

- Hold each gesture steady — don't rush the hold
- Transition quickly between gestures (the pause is the fast part)
- Verify each word appears before moving to the next

Exercise 5: Letter Spelling

Switch to **Letters mode** and spell: **P-A-I-N**

Then switch back to **Words mode** and continue with gestures.

> 💡 **Training Goal:** Staff should be able to demonstrate all 15 gestures correctly and explain each department's unique word mappings to patients. Patients should be able to build a 5-word sentence within 30 seconds.

© 2026 — For training purposes only